Royal Devon and Exeter NHS
NHS Foundation Trust

# Clinical Guideline for:
# Management of collapse / transient loss of consciousness in children and young people (TLOC)

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| **Version number** | 4.0 |
| **Replaces version number** | 3.0 |
| **Consultation undertaken with:** | Paediatric Governance Group |
| **Approving body and date approved** | Paediatric Governance Group 08/12/2021 |
| **Review date** | 08/09/2026 (3-6 months prior to expiry date) |
| **Expiry date** | 08/12/2026 |
| **Date document becomes live** | 09/12/2021 |

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## 1. BACKGROUND AND SCOPE OF THIS GUIDELINE

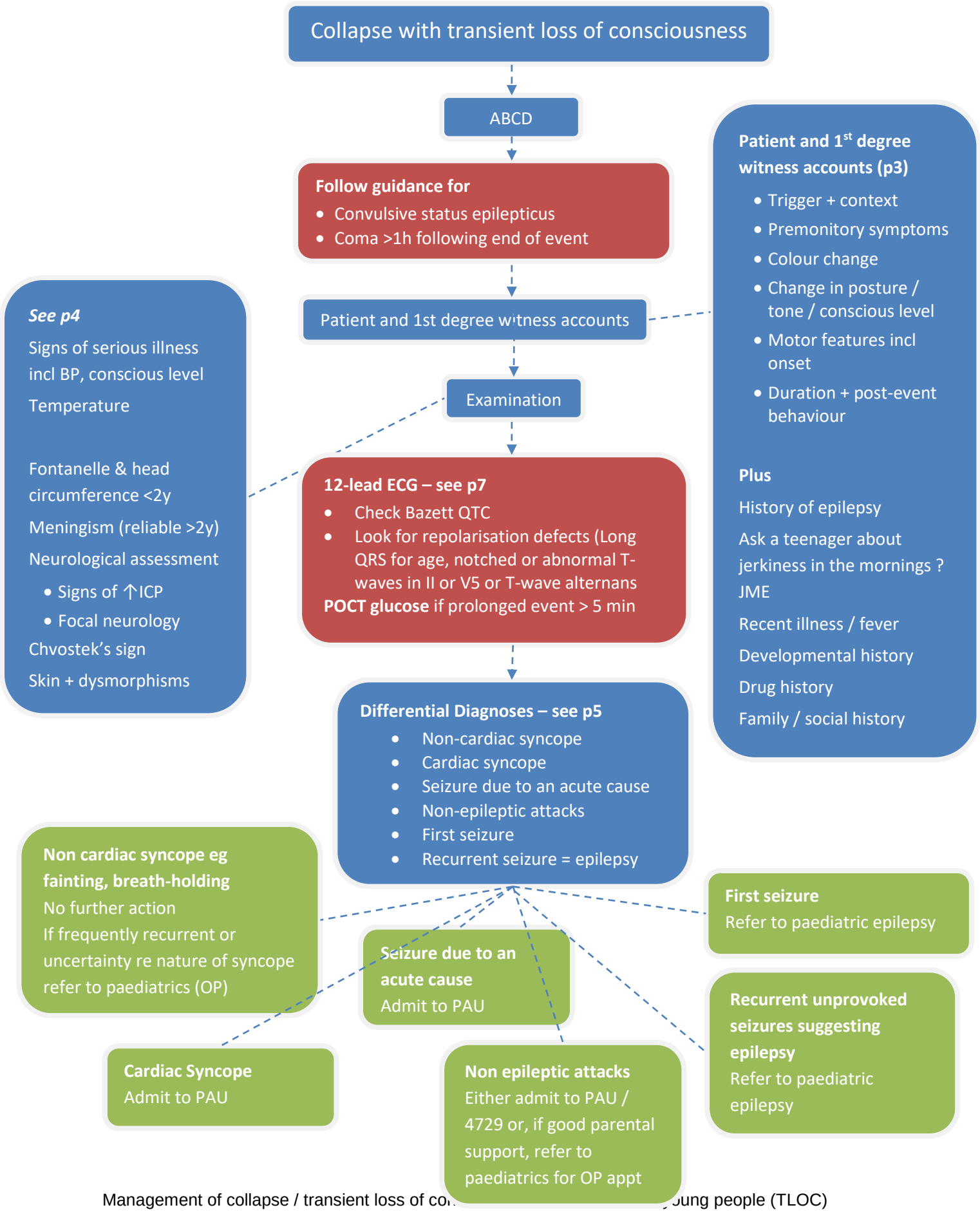
1.1 Most events associated with collapse, transient loss of consciousness or twitching do not represent an epileptic seizure. Over diagnosis in epilepsy is common and associated with harmful consequences eg anti-epileptic medication side-effects and the impact of the diagnosis on school and social life. Many important differential diagnoses are commonly misdiagnosed as epilepsy, including potentially fatal long QT syndromes.

1.2 This guideline is intended for use with children or adolescents <18 years presenting with an episode of collapse / TLOC. It includes details of the differential diagnoses so that correct advice and follow-up is given and only appropriate investigations undertaken.

1.3 This guideline does not cover children or young people with recurrent blank spells.

1.4 This guideline may not be appropriate in all circumstances. If in doubt discuss with a senior doctor.

## 2. ALGORITHIM



## 3. INITIAL MANAGEMENT

3.1 Assess ± stabilise ABC at presentation to the emergency department.

- Convulsive seizure lasting more than 5 minutes → manage in resus as per convulsive status epilepticus guideline.
- Suspected seizure in a child <6y associated with a fever → manage as per febrile seizure guideline.
- Any child / adolescent with decreased conscious level preceding a seizure, or lasting for >1h following a seizure → manage as per decreased conscious level guideline.
- Any child / young person with known epilepsy or other cause for collapse → manage as per EPIC 'FYI' emergency treatment plan / patient held seizure management plan where these exist.

## 4. HISTORY

4.1 Get a detailed history of the event from the patient and first-hand witness. Do this even where a diagnosis of epilepsy has already been made – it may be wrong.

4.2 History of the event itself

- Triggers / context in which the event happened – sleep, exercise, standing up from seating, anxiety, flashing lights
- Symptoms before the collapse / seizure started (ask the patient about this) – hearing or visual change, dizziness, epigastric pain, psychic symptoms
- Change in colour before / during event
- Change in posture / tone – stiff or floppy when collapsing? eye / head deviation?

- Change in conscious level
- Onset of event itself – focal posturing?
- Motor phenomena – shivering, jerking, flailing? Ask the witness to demonstrate for you
- Other features during event – breathing difficulty / change? Tongue biting? Incontinence? (NB latter two are poor predictors of seizures)
- Duration of event
- Symptoms following event – post-ictal behaviour is uniquely associated with confusion

The best discriminators between vasovagal syncope and a seizure are (1) presence of premonitory features such as light-headedness, visual loss and (2) pallor seen in syncope but rare in seizures.

4.3 Other features in the history

- Previous diagnosis of epilepsy
- Ask a teenager with TLOC whether they experience myoclonic jerks in the mornings, causing them to drop breakfast crockery or mess up make-up application. This might signify myoclonic jerks, seen with TLOC in juvenile myoclonic epilepsy (JME).
- Recent health – intercurrent illness, fever, other new neurological symptoms including change in personality / developmental regression
- Known or suspected childhood epilepsy - ask about similarity to previous seizures, usual seizure control, compliance to treatment
- Developmental history including education
- Drug history (ask about compliance with AED treatment for patients with epilepsy)
- Family history – sudden death with exercise / sleep, congenital deafness, epilepsy

## 5. EXAMINATION

5.1 Assess the following in all children / adolescents with collapse / seizure

- Signs of serious illness, including HR, blood pressure, conscious level assessment
- Temperature
- Fontanelle & head circumference <2y
- Meningism (Kernig + Brudzinski signs more accurate than 'neck stiffness', but all are poor predictors of meningitis in children <2 years)
- Neurological assessment – look for signs of raised intracranial pressure / focal neurology.
- Chvostek's sign (ipsilateral facial contraction on percussion of facial nerve = hypocalcaemia)

- Skin / general appearance (neurocutaneous or other dysmorphic syndromes)

## 6. INVESTIGATIONS

### 6.1 12-lead ECG

**12-lead ECG**

The single essential test in every case of first collapse / seizure[1]

Verify the ECG's automated QTc with a manual estimation of QTc using Bazett's formula.

Look for signs of repolarisation problems – wide QRS for age, notched T-waves or flipping T-waves best seen on the rhythm strip on the bottom of the ECG. See Appendix 1 (page 7)

### 6.2 Blood glucose

**Blood glucose**

Only perform a bedside blood glucose measurement if the patient is still seizing or is not fully alert.

***Do not 'routinely' order***

**EEG** – not required following a first seizure. Used only to support epilepsy classification or guide treatment and prognostication in epilepsy. Do not order to 'rule out' epilepsy or to predict seizure recurrence. Discuss with one of the paediatric epilepsy team first if you are unsure whether this is indicated.

**Other blood tests** (Ca, Mg, U&E, FBC) – not usually required, but order in children <18 months, those with reduced conscious level or where the history / examination suggest an acute cause

**Urine toxicology** – not usually required, but order if failure to recover as expected from a TLOC or coma

**Brain imaging** – do not order just because of a suspected focal seizure. Order where you suspect an acute cause of an epileptic seizure eg traumatic brain injury or stroke. CT good for fresh blood or stroke. Most first seizures occur in people with normal brain scans

[1] NICE CG20 *Diagnosis + management of the epilepsies in children* updated 2020 @ Overview | Epilepsies: diagnosis and management | Guidance | NICE

## 7. FURTHER MANAGEMENT DEPENDS ON CAUSE OF TLOC

7.1 If you are uncertain about the cause, discuss with one of the paediatric epilepsy team. If out of hours discuss with a senior paediatrician on the acute team.

**Non cardiac syncope eg fainting, breath-holding**

- No further action
- If frequently recurrent or uncertainty re nature of syncope refer to paediatrics
- Give advice and patient information *(see Appendix 2 for links)*

- Vasovagal syncope (fainting) - by far the commonest cause of syncope in the young. One in three teenagers faint at least once; many more have pre-sycnopal symptoms without blacking out. 50% patients go stiff and jerk during fainting episodes.

- Reflex anoxic seizures ('pallid breath holding attacks') and 'blue' breath holding attacks – very anxiety provoking by parents / teachers, but not dangerous. The clue is the triggers: a fright or a bumped head, vigorous crying. No further action required unless very frequent.

**Cardiac Syncope**

- Admit to PAU for senior paediatric review
- Next test depends on precise cause but likely to be 24-hour tape

- This is very rare but can be fatal
- More commonly caused by drugs lengthening QT interval (eg Amitriptyline in overdose, combinations of medications such as Carbamazepine, Domperidone), but also genetic long QT syndrome and structural heart lesions
- Ensure you have checked the QT interval on 12-lead ECG, taken a drug history and asked about family members with syncope / sudden death

**Non-epileptic attacks**

- Either admit to paediatrics or, if good parental support, refer to 'paediatrics' for OP apt
- If admitted, for senior paediatric review. Get a psychosocial history
- Good online resources – www.nonepilepticattacks.info + www.neurosymptoms.org.uk

- This is a functional neurological disorder. This is usually a clinical diagnosis. Though not necessary, video-EEG may help to secure the diagnosis where you are uncertain and / or get family buy-in.
- Discriminating features as per the table on the following page.

| | **Tonic-clonic seizure** | **Syncope** | **Non-epileptic attack** |
|---|---|---|---|
| **Precipitant** | Variable, sleep onset (any age), sleep deprivation in adolescents | Pain, getting up quickly after prolonged sitting<br>Never lying down | Variable, usually in company |
| **Premonitory Symptoms** | Rarely identified | Dizziness, sweating, nausea, visual and hearing changes | Variable |
| **Duration** | Minutes | Seconds | Minutes – longest of all |
| **Eyes** | Open | Open | Often tightly shut |
| **Rigidity** | + | +/- | Never sustained |
| **Clonic movements** | +/- | +/- | Flailing, not jerks<br>Variable intensity over time<br>No facial twitching |
| **Colour change** | Variable | Marked pallor | - |

| | | | |
|---|---|---|---|
| **Breathing change** | Stertor / “snoring” | - | Moans. Crying at end |
| **Self-injury** | Front or side of tongue | +/- | Front of tongue |
| **Incontinence** | + | + | + |
| **Post-ictal confusion** | + | - | Can be tearful |

**Seizure due to an acute cause**

- Manage in resus or admit to PAU
- Further investigation & treatment depends on cause

Other than febrile seizures, these are uncommon in isolation. Potential causes are metabolic (commonest – low Na, glucose), toxic (accidental ingestion, self harm with drugs such as Mefanamic acid, or poisoning), a bleed including NAHI in young children, a stroke, encephalitis or ADEM. History and examination almost always provide other clues.

Due to the presence of clues in the history or examination, there is no place for a general ‘routine’ order set of investigations / treatment for all children with seizures in whom an acute cause is suspected. Instead, consider the following:

- **For features of self-harm / poisoning** (eg dilated pupils, sweating, ataxia, drowsiness) – urine toxicology, find out what is at home / on the patient’s possession & search Toxbase
- **For features of metabolic disorder** (eg shock, drowsiness) – blood gas, glucose, blood ketones, bicarbonate, renal + liver function, calculate anion gap
- **For suspected infection** – FBC, cultures, CRP, CSF analysis. Lumbar puncture is safe where GCS is 13 or above, in a child <2y and an hour following a seizure has ended.
- **For suspected ADEM** – CT / MRI +CSF analysis
- **For suspected bleed / trauma / stroke*–** CT head (*seek advice re timing after a suspected stroke)

**First seizure**

- Refer to paediatric epilepsy
- Will be triaged quickly with OP within 2-3 weeks & telephone call from Penny Smith, epilepsy specialist nurse
- Provide first aid + safety advice / links especially in relation to bathing (see Appendix 2)

One in 20 children have a single convulsive seizure before the age of 16. Only 40% will go on to have more within 2 years. A child / young person is more likely to have more seizures (and therefore epilepsy) if they have another neurological condition eg cerebral palsy.

Do not 'routinely' order an EEG after a first seizure – this provides very limited predictive value. 'False positive' findings are common especially in children with other neurological conditions or a family history of epilepsy.

If you are uncertain discuss the patient with a member of the paediatric epilepsy team first – see contact details in Appendix 3.

**Recurrent unprovoked seizures**

- Refer to paediatric epilepsy
- Will be triaged quickly with OP within 2-3 weeks & telephone call from Penny Smith, epilepsy specialist nurse
- Provide first aid + safety advice / links especially in relation to bathing *(see Appendix 2)*

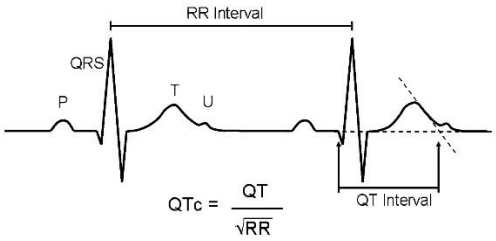
Apart from headache, epilepsy is the commonest neurological condition in children and young people. Childhood epilepsy syndromes include a wide spectrum of disorders ranging from being benign to life threatening. There are many known epilepsy syndromes and potential contributory factors such as hypoxic brain injury before or after birth, CNS infection or traumatic brain injury. However, ~ 40% of childhood epilepsies are thought to have some component of genetic involvement. New genes, mutations, and variants involved in epilepsy are being identified continuously.

Children and young people with epilepsy in the Exeter / Mid + East Devon area are cared for by a specialist paediatric epilepsy team to whom you should refer children and young people who have recurrent seizures. The team contact details are in Appendix 3.

## APPENDIX 1: USE OF 12-LEAD ECG TO DETERMINE QTC

(1) Manually estimate the QTc using Bazett's formula .

- Using leads II or V5 extrapolate the steepest part of the descent of the T-wave to where it intersects the baseline: this is the end of the T-wave
- U-waves are common in adolescents. Include a U-wave in the T-wave measurement if > 50% of the height of the T wave. Exclude the U-wave if it is separate from / smaller than the T-wave.
- Measure the RR and QT intervals




(2) Look for repolarisation defects associated with LQTS:

- Long QRS for age
- Notched or abnormal T-waves in II or V5
- T-wave alternans (below)

(3) Always consider the QTc in the context of the patient's personal and family history:

- Syncope / seizure with exercise (especially swimming) or associated with emotion or sleep
- Family history of LQTS, sudden death in a 1st degree relative < 30yrs or congenital deafness

**Interpretation**
1. The most important indicators of LQTS are the context of the collapse plus family history
2. The normal range for QTc is 350 – 450ms (95th percentile limits). A QTc > 0.47 seconds is suggestive of a long QT syndrome. QTc > 0.44 seconds is 'borderline'.

**Where QT is suggestive of LQTS**
1. Order 24 hour ECG - ask for QT interval measurement in comments box
2. Refer to 'paediatric cardiology' urgently and contact one of the local consultants (Drs Mabin / Osborne / Venkata).
3. If not available locally, refer to paediatric cardiology in Bristol Children's Hospital. You will be asked to send a copy of the ECG.
4. Order exercise test if recommended by paediatric cardiologist. Children < 15 years with syncope are usually exercise tested in Bristol Children's. Over 16 years may be tested in Exeter, but need Adult cardiology SpR supervision

**Where QTc is 'borderline' and there are no predictors in the patient or family history of LQTS**
- Refer to 'paediatric cardiology'
- Order 24 hour ECG - ask for QT interval measurement in comments box

## APPENDIX 2: USEFUL RESOURCES FOR PATIENTS/PARENTS

- **Vasovagal syncope**

Reflex Syncope (Vasovagal Syncope) - Arrhythmia Alliance Group (heartrhythmalliance.org)
Fainting - NHS (www.nhs.uk)

- **Reflex Anoxic Seizures**

Reflexic Anoxic Seizures. What are Seizures? Information page | Patient
Reflex Anoxic Seizures (RAS) - Arrhythmia Alliance Group (heartrhythmalliance.org)

- **Breath-holding attacks**

Breath-holding in babies and children - NHS (www.nhs.uk)

- **First & recurrent seizure/s**

Local parent information sheet available on paediatric pages of IaN..\First Seizure Info Sheet June 2011.pub

First aid for focal and bilateral tonic-clonic seizures available at
What to do when someone has a seizure | Epilepsy Action

Epilepsy Action (www.epilepsy.org.uk) is an excellent resource for first aid (see above) and safety advice. Advise to take showers or always be supervised in the bath. Advise driving teenagers to stop driving.

## APPENDIX 3 – PAEDIATRIC EPILEPSY TEAM

| | | |
|---|---|---|
| Penny Smith | Epilepsy Specialist Nurse | Pennysmith3@nhs.net<br>07554113801 / 01392-406547 |
| Helen Brewer | Consultant Paediatrician | Helen.brewer@nhs.net<br>07958134754 |
| Rachel Howells | Consultant Paediatrician | Rachel.howells@nhs.net<br>01392-402668 |
| Leighton Phillips | Consultant Paediatrician | Leighton.phillips1@nhs.net |
| Eleanor Thomas | Consultant Paediatrician | Eleanor.thomas4@nhs.net<br>07788580127 |
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